Bellman–Ford Algorithm

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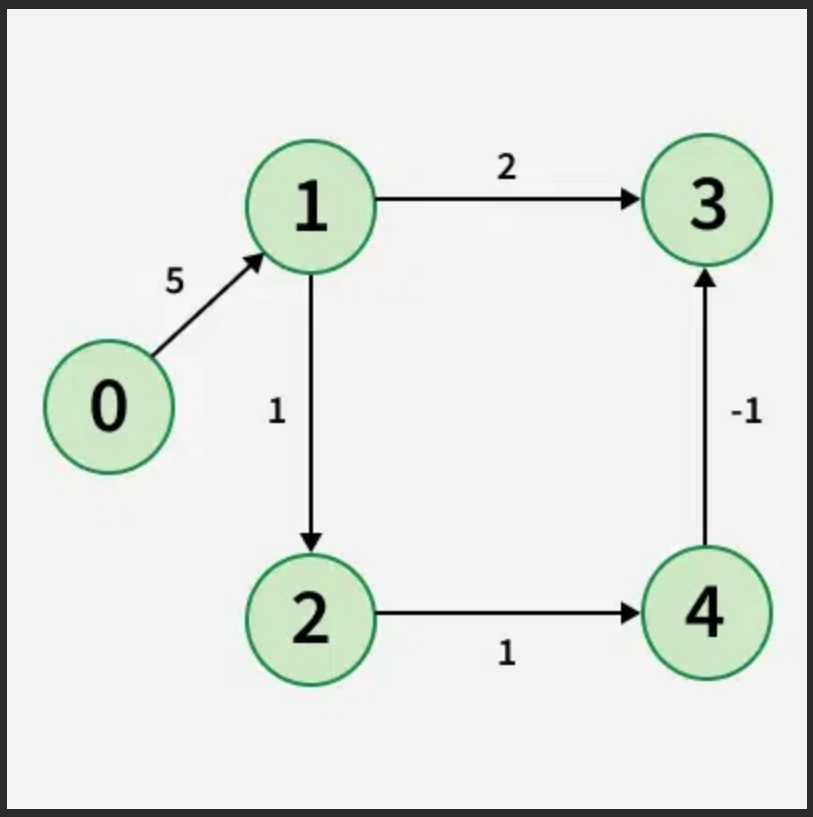
Given a weighted graph with **V** vertices and **E** edges, along with a source vertex `src`, the task is to compute the shortest distances from the source to all other vertices. If a vertex is unreachable from the source, its distance should be marked as $\mathbf{10^8}$. In the presence of a negative weight cycle, **return -1** to signify that shortest path calculations are not feasible.

Examples:

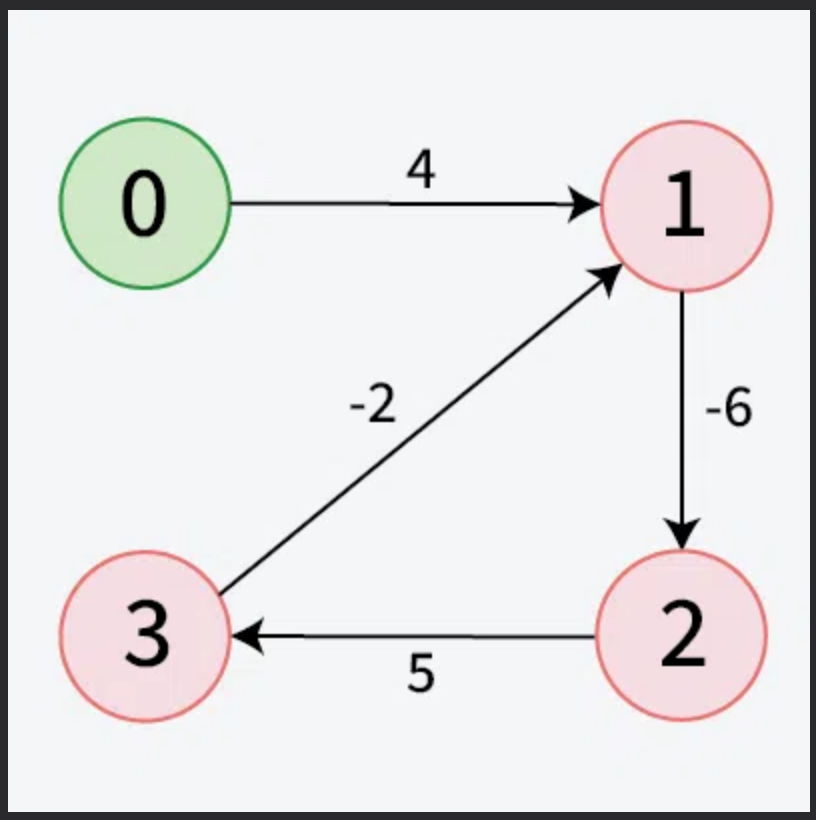
Input: *V = 5, edges = [[0, 1, 5], [1, 2, 1], [1, 3, 2], [2, 4, 1], [4, 3, -1]], src = 0*



Output: *[0, 5, 6, 6, 7]*
Explanation: *Shortest Paths:*
For 0 to 1 minimum distance will be 5. By following path 0 → 1

For 0 to 2 minimum distance will be 6. By following path 0 → 1 → 2
For 0 to 3 minimum distance will be 6. By following path 0 → 1 → 2 → 4 → 3
For 0 to 4 minimum distance will be 7. By following path 0 → 1 → 2 → 4

***Input**: V = 4, edges = [[0, 1, 4], [1, 2, -6], [2, 3, 5], [3, 1, -2]], src = 0*



***Output**: [-1]*
***Explanation:** The graph contains a negative weight cycle formed by the path **1** → **2** → **3** → **1**, where the total weight of the cycle is negative.*

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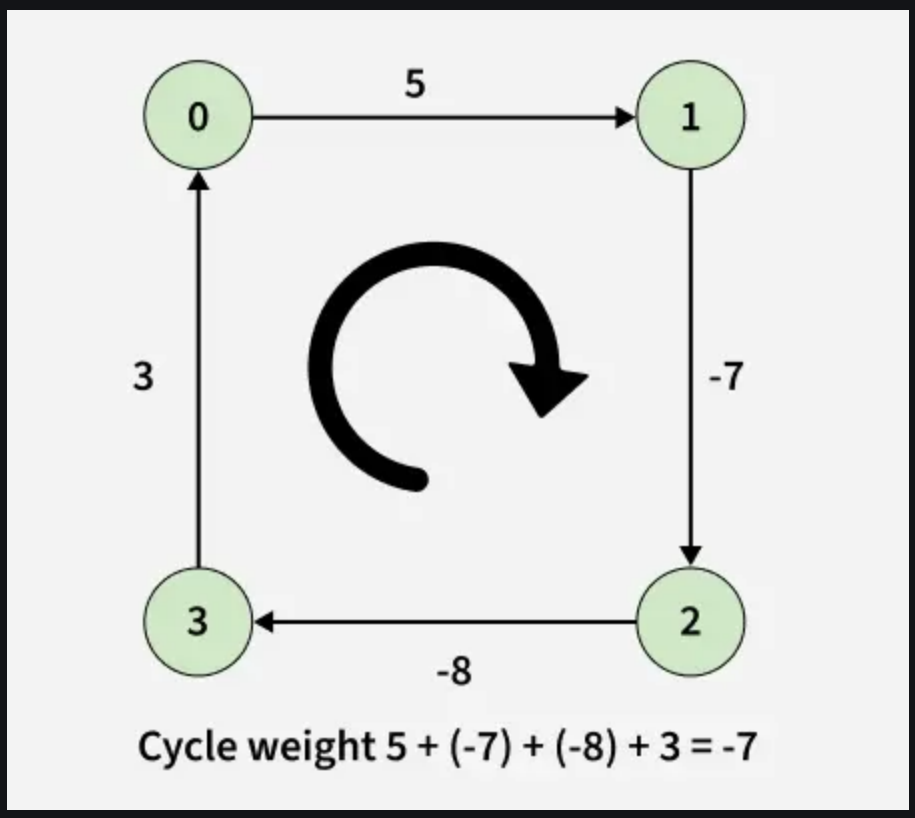
Approach: Bellman-Ford Algorithm - O(V*E) Time and O(V) Space

Negative weight cycle:

A negative weight cycle is a cycle in a graph, whose sum of edge weights is negative. If you traverse the cycle, the total weight accumulated would be less than zero.



In the presence of negative weight cycle in the graph, the shortest path **doesn't exist** because with each traversal of the cycle shortest path keeps **decreasing**.

Limitation of Dijkstra's Algorithm:

Since, we need to find the **single source shortest path**, we might initially think of using Dijkstra's algorithm. However, Dijkstra is not suitable when the graph consists of **negative edges**. The reason is, it **doesn't revisit**

those nodes which have already been marked as visited. If a shorter path exists through a longer route with negative edges, Dijkstra's algorithm will fail to handle it.

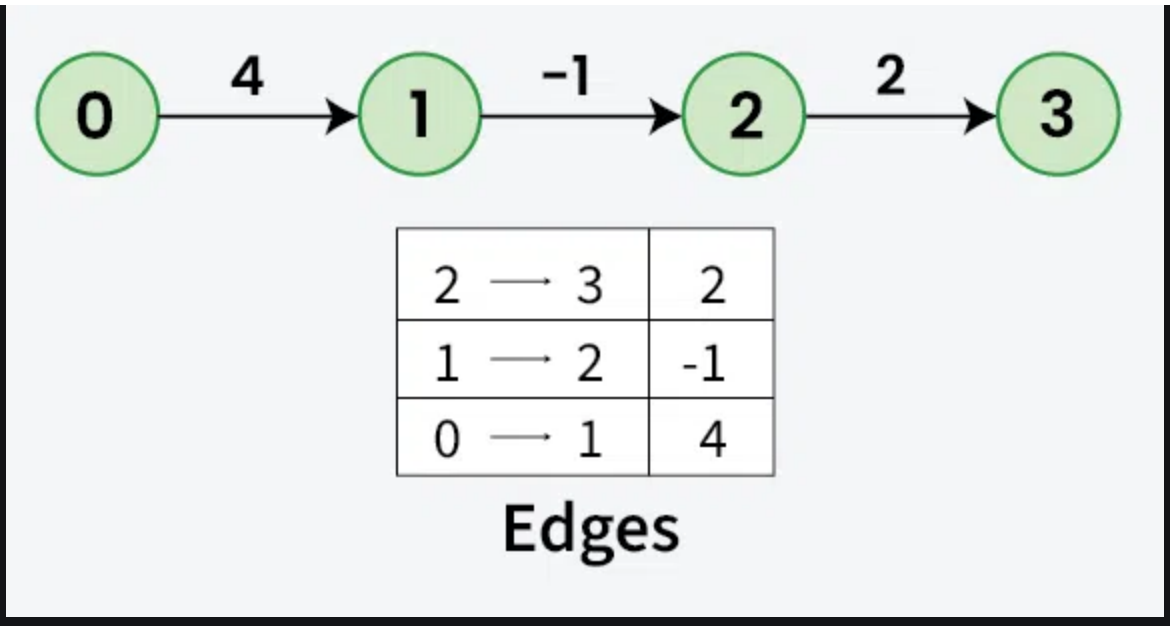
Principle of Relaxation of Edges

- Relaxation means **updating** the shortest distance to a node if a shorter path is found through another node. For an edge **(u, v)** with weight **w**:
  - If going through u gives a shorter path to v from the source node (i.e., **distance[v] > distance[u] + w**), we update the distance[v] as distance[u] + w.
- In the bellman-ford algorithm, this process is repeated **(V - 1)** times for all the edges.

Why Relaxing Edges (V - 1) times gives us Single Source Shortest Path?

A shortest path between two vertices can have at most **(V - 1)** edges. It is not possible to have a simple path with more than (V - 1) edges (otherwise it would form a cycle). Therefore, repeating the relaxation process (V - 1) times ensures that all possible paths between source and any other node have been covered.

Assuming node **0** as the **source** vertex, let's see how we can relax the edges to find the shortest paths:



- In the **first relaxation**, since the shortest paths for vertices 1 and 2 are unknown (infinite, i.e., 10^8), the shortest paths for vertices 2 and 3 will also remain infinite (10^8) . And, for vertex 1, the distance will be updated to 4, as `dist[0] + 4 < dist[1]` (i.e., `0 + 4 <` 10^8).
- In the **second relaxation**, the shortest path for vertex 2 is still infinite (e.g. 10^8), which means the shortest path for vertex 3 will also remain infinite. For vertex 2, the distance can be updated to 3, as dist[1] + (-1) = 3.
- In the **third relaxation**, the shortest path for vertex 3 will be updated to 5, as dist[2] + 2 = 5.

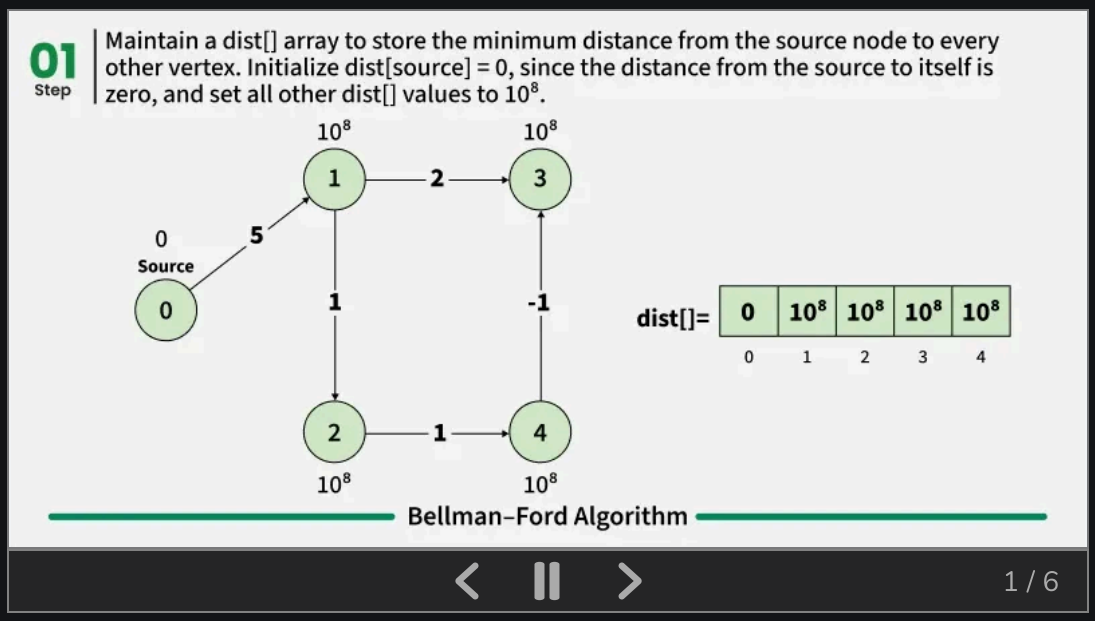
So, in above example, **dist[1]** is updated in **1st relaxation**, **dist[2]** is updated in **second relaxation**, so the dist for the last node (V - 1), will be updated in **(V - 1) th relaxation**.

Detection of a Negative Weight Cycle

- As we have discussed earlier that, we need (V - 1) relaxations of all the edges to achieve single source shortest path. If one **additional relaxation** (Vth) for any edge is possible, it indicates that some edges with overall negative weight has been traversed once more. This indicates the presence of a **negative weight cycle** in the graph.

***Bellman-Ford** is a **single source** shortest path algorithm. It effectively works in the cases of negative edges and is able to detect negative cycles as well. It works on the principle of **relaxation of the edges**.*

Illustration:



C++ Java Python C# JavaScript

```cpp
#include <iostream>
#include <vector>
using namespace std;

vector<int> bellmanFord(int V, vector<vector<int>>& edges, int src)
{

    // Initially distance from source to all
    // other vertices is not known(Infinite).
    vector<int> dist(V, 1e8);
    dist[src] = 0;

    // Relaxation of all the edges V times, not (V - 1) as we
    // need one additional relaxation to detect negative cycle
    for (int i = 0; i < V; i++) {
```

```
        for (vector<int> edge : edges) {
            int u = edge[0];
            int v = edge[1];
            int wt = edge[2];
            if (dist[u] != 1e8 && dist[u] + wt < dist[v]) {

                // If this is the Vth relaxation, then there is
                // a negative cycle
                if(i == V - 1)
                    return {-1};

                // Update shortest distance to node v
                dist[v] = dist[u] + wt;
            }
        }
    }

    return dist;
}

int main() {

    // Number of vertices in the graph
    int V = 5;

    // Edge list representation: {source, destination, weight}
    vector<vector<int>> edges = {
        {1, 3, 2},
        {4, 3, -1},
        {2, 4, 1},
        {1, 2, 1},
        {0, 1, 5}
    };

    // Define the source vertex
    int src = 0;

    // Run Bellman-Ford algorithm to get shortest paths from src
    vector<int> ans = bellmanFord(V, edges, src);
```

```
    // Output the shortest distances from src to all vertices
    for (int dist : ans)
        cout << dist << " ";

    return 0;
}
```

Output

```
0 5 6 6 7
```

Problems based on Shortest Path

- Shortest Path in Directed Acyclic Graph
- Shortest path with one curved edge in an undirected Graph
- Minimum Cost Path
- Path with smallest difference between consecutive cells
- Print negative weight cycle in a Directed Graph
- 1st to Kth shortest path lengths in given Graph
- Shortest path in a Binary Maze
- Minimum steps to reach target by a Knight
- Number of ways to reach at destination in shortest time
- Snake and Ladder Problem
- Word Ladder

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